



M2 Conservation of Momentum LAB Report

An Introduction To Engineering & Practices (Nanyang Technological University)



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NANYANG TECHNOLOGICAL UNIVERSITY
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**M2 CONSERVATION OF MOMENTUM
LAB REPORT**

Mechanics of Materials (MAE)
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ABSTRACT

This report investigates the principle of conservation of momentum in both elastic and inelastic collisions using a nearly frictionless air track. Sliders of equal and unequal masses were collided and the velocities before and after the collisions were measured using photo-gate timers. The results confirmed that the total momentum remained constant in both types of collisions, regardless of mass differences. Additionally, it was observed that kinetic energy was conserved in elastic collisions but not in inelastic collisions, where some energy was lost, likely as heat, sound or deformation. The experiment successfully demonstrated the application of momentum conservation laws and highlighted differences in energy behavior between elastic and inelastic collisions.



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M2 CONSERVATION OF MOMENTUM

1. INTRODUCTION

The principle of conservation of momentum is fundamental in mechanics, stating that the total momentum of a closed system remains constant if no external forces act upon it. This experiment aims to verify this principle by studying collisions between two sliders on a nearly frictionless air track. Collisions are classified into two types: elastic, where both momentum and kinetic energy are conserved, and inelastic, where only momentum is conserved while some kinetic energy is lost to other forms such as heat, sound or deformation.

In this experiment, two scenarios were examined: (1) elastic collisions where the sliders rebound from each other, and (2) inelastic collisions where the sliders stick together post-collision. Through precise measurements of velocity before and after the collisions, this experiment seeks to demonstrate that the total momentum of the system remains conserved in both types of collisions, providing experimental verification of the conservation of momentum.

2. EQUIPMENT USED

The experiment was conducted using a nearly frictionless air track connected to an air supply, which minimized external forces such as friction. Two sliders were used, each equipped with bumpers designed for elastic and inelastic collisions. Digital photo-gate timers were placed along the track to accurately measure the time taken by each slider to pass the sensors before and after the collisions, allowing for velocity calculations. Additional equipment included a ruler for measuring the length of the sliders, ensuring consistency in the analysis. This setup allowed for precise observation of the collision dynamics while minimizing external interferences.

3. BACKGROUND THEORY

The principle of conservation of momentum states that the total momentum of a system remains constant if no external forces act on it. In collisions, the momentum before and after the collision should be equal for a closed system. For two colliding bodies, the total momentum before the collision is given by:

$$P_{\text{initial}} = M_1 U_1 + M_2 U_2$$

where:

- M_1 and M_2 are the masses of the two bodies,
- U_1 and U_2 are their initial velocities.

After the collision, the total momentum is:

$$P_{\text{final}} = M_1 V_1 + M_2 V_2$$

where V_1 and V_2 are the velocities of the two bodies after the collision. For momentum to be conserved, the total momentum before and after the collision must be equal:



$$M_1U_1 + M_2U_2 = M_1V_1 + M_2V_2$$

In an elastic collision, both momentum and kinetic energy are conserved. The total kinetic energy of the system before and after the collision is described by:

$$KE_{\text{initial}} = \frac{1}{2}M_1U_1^2 + \frac{1}{2}M_2U_2^2$$

and after the collision:

$$KE_{\text{final}} = \frac{1}{2}M_1V_1^2 + \frac{1}{2}M_2V_2^2$$

In inelastic collisions, while momentum is still conserved, kinetic energy is not fully conserved due to energy loss in the form of heat, sound, or deformation. Thus, the total kinetic energy after an inelastic collision is less than before the collision.

This experiment explores both elastic and inelastic collisions to verify that, regardless of whether kinetic energy is conserved, momentum remains constant as long as no external forces interfere. Additionally, the coefficient of restitution, denoted by e , is a measure of how elastic or inelastic a collision is. It is defined as the ratio of the relative velocity of separation to the relative velocity of approach:

$$e = \frac{V_2 - V_1}{U_1 - U_2}$$

- For elastic collisions, $e = 1$, indicating no kinetic energy is lost.
- For completely inelastic collisions, $e = 0$, meaning the objects stick together and move with the same velocity after the collision.

This coefficient provides insight into the elasticity of the collision, which, when combined with the conservation laws of momentum and kinetic energy, helps classify the nature of the interaction between the colliding bodies.

4. EXPERIMENTAL PROCEDURE

The experiment was conducted to observe the conservation of momentum and kinetic energy in both elastic and inelastic collisions using two sliders on a nearly frictionless air track. The air track was set up horizontally to minimize external forces, such as friction, ensuring that the measurements accurately reflected the interactions between the sliders. Digital photo-gate timers were strategically placed along the track to measure the time intervals as the sliders passed through the sensors, enabling precise calculations of their velocities before and after collisions. The setup facilitated the examination of how momentum and kinetic energy were conserved or transformed during different types of collisions.



The following steps were carried out:

1. **Setup of the Equipment:** The air track was connected to the air supply and the sliders were equipped with bumpers suitable for elastic and inelastic collisions. The track was levelled to ensure that the sliders would remain stationary unless pushed. The photo-gate timers were switched on and their sensors were positioned at two points on the track to record the passage of the sliders before and after collisions.
2. **Elastic Collisions:** For the first part of the experiment, two sliders of equal mass were used. Slider 1 was placed at the start of the track and slider 2 was positioned at rest between the sensors. Slider 1 was pushed to collide with slider 2 and the time intervals were recorded by the sensors before and after the collision. The velocity of each slider was calculated using the measured time and the length of the sliders. This procedure was repeated multiple times with different initial velocities of slider 1.
3. **Elastic Collisions with Unequal Masses:** For the second part, the mass of slider 2 was increased to approximately twice that of slider 1. Slider 1 was again placed at the start of the track and slider 2 was positioned at rest. Slider 1 was pushed to collide with slider 2 and the times were recorded as before. This was repeated with varying initial velocities of slider 1 to observe how the differences in mass affected the results.
4. **Inelastic Collisions with Unequal Masses:** For the final part of the experiment, the sliders were configured for inelastic collisions using special bumpers that allowed them to stick together after colliding. Slider 1 was pushed to collide with slider 2 and the velocities of the combined sliders after the collision were measured. The process was repeated with different initial velocities of slider 1 to gather sufficient data for analysis.
Throughout the experiment, care was taken to switch off the air supply when the track was not in use, to minimize noise and unnecessary air flow. After collecting the data, the velocities of the sliders before and after each collision were calculated and recorded for further analysis.

5. EXPERIMENTAL RESULTS

5.1. ELASTIC COLLISIONS OF SLIDERS HAVING THE SAME MASS

mass of slider 1 = 0.3 kg
length of slider 1 = 0.18 m

mass of slider 2 = 0.3 kg
length of slider 2 = 0.18 m



t_A (s)	$\hat{t}_B$ (s)	u_1 (m/s)	v_2 (m/s)
0.1805	0.2015	0.997	0.893
0.184	0.2082	0.978	0.865
0.269	0.2931	0.669	0.614

Table 1. Results for elastic collisions of sliders having the same mass.

5.2. ELASTIC COLLISION OF SLIDERS HAVING DIFFERENT MASSES

mass of slider 1 = 0.3 kg
length of slider 1 = 0.18 m

mass of slider 2 = 0.45 kg
length of slider 2 = 0.29 m

t_A (s)	$\hat{t}_A$ (s)	$\hat{t}_B$ (s)	u_1 (m/s)	v_1 (m/s)	v_2 (m/s)
0.5581	2.1729	1.3399	0.323	-0.0828	0.216
0.3226	1.7619	0.6854	0.558	-0.102	0.423
0.2399	1.2820	0.5091	0.750	-0.140	0.570

Table 2. Results for elastic collisions of sliders having different masses.

p (Ns)	$\hat{p}$ (Ns)	$\frac{p - \hat{p}}{p} \times 100\%$	E (Nm)	$\hat{E}$ (Nm)	$\frac{E - \hat{E}}{E} \times 100\%$
0.0969	0.07236	25.325	0.0156	0.0115	26.282
0.1674	0.15975	4.570	0.0467	0.0418	10.493
0.225	0.2145	4.667	0.0844	0.0760	9.953

Table 3. Total momentum and kinetic energy before and after collision.

5.3. INELASTIC COLLISIONS OF SLIDERS HAVING DIFFERENT MASSES

mass of slider 1 = 0.3 kg
length of slider 1 = 0.18 m

mass of slider 2 = 0.45 kg
length of slider 2 = 0.29 m

t_A (s)	$\hat{t}_B$ (s)	u_1 (ms ⁻¹)	$v_1 = v_2$ (ms ⁻¹)
0.5237	2.3483	0.3437	0.1235
0.4424	1.8390	0.40687	0.1577
0.6093	2.8768	0.29542	0.1008

Table 4. Inelastic collisions of sliders having different mass.



p (Ns)	$\hat{p}$ (Ns)	$\frac{p - \hat{p}}{p} \times 100\%$	E (Nm)	$\hat{E}$ (Nm)	$\frac{E - \hat{E}}{E} \times 100\%$
0.1031	0.09263	10.16	0.01772	0.005720	67.72
0.1221	0.118275	3.13	0.02483	0.009326	62.44
0.08863	0.0756	14.70	0.01309	0.0038102	70.9

Table 5. Total momentum and kinetic energy before and after collision.

6. DISCUSSION

6.1. ELASTIC COLLISIONS OF SLIDERS HAVING THE SAME MASS

For elastic collisions between sliders of equal mass, it was observed that slider 1's velocity significantly decreased after colliding with slider 2. This reduction in velocity occurs because most of slider 1's kinetic energy is transferred to slider 2. As a result, slider 2, initially at rest, begins to move, while slider 1 experiences a notable decrease in velocity. This behavior is consistent with the expectation that kinetic energy and momentum are transferred between the two bodies during elastic collisions.

6.2. MOMENTUM TRANSFER

In elastic collisions, the momentum transfer was more efficient when slider 1 moved at higher velocities. The experimental data showed an average percentage error of 4.5% under these conditions, indicating that the initial momentum of slider 1 closely matched the combined final momentum of both sliders after the collision. When slider 1 moved off at a slower velocity, a higher percentage error was observed, likely due to increased air resistance acting on the slider. For inelastic collisions, momentum transfer was less efficient, with an average percentage error of 10%. This suggests that the initial momentum of the system was higher than the final momentum, likely due to additional losses caused by factors such as friction and deformation.

6.3. ENERGY TRANSFER

For energy transfer in elastic collisions, it was noted that when slider 1 moved more slowly, the percentage error increased, likely due to the longer time for energy to be lost to the surroundings through heat or other forms. At higher velocities, the energy transfer was more efficient, with an average percentage error of 10%. However, in inelastic collisions, the energy loss was significantly greater, with an average error of 65%. The higher energy loss in inelastic collisions can be attributed to the conversion of kinetic energy into heat and



sound, as well as frictional forces between the sliders, which increased when they stuck together after the collision.

6.4. COEFFICIENT OF RESTITUTION

The calculated average coefficient of restitution for the elastic collisions was 0.94. This value is reasonable as it is close to the ideal value of 1 for perfectly elastic collisions. The small deviation from 1 can be attributed to experimental imperfections such as air resistance, noise and slight energy losses. Despite these factors, the results demonstrate that the collisions were nearly elastic.

6.5. SOURCES OF ERROR

Several sources of error were identified in this experiment. First, the use of two photo-gate timers may have introduced timing discrepancies, as it is challenging to ensure both timers are perfectly synchronized. This discrepancy could result in an estimated percentage error of 2%. Additionally, external factors such as air resistance and noise likely contributed to an error of approximately 20%, particularly in affecting the velocities of the sliders and introducing variability in the results.

6.6. CONSERVATION OF MOMENTUM AND NEWTON'S LAWS

The principle of conservation of momentum is derived from Newton's second law of motion. Newton's second law states that the momentum of an object is the product of its mass and velocity, and the rate of change of momentum is proportional to the net external force acting on the object. In the absence of external forces, the total momentum of a system remains constant. Thus, the law of conservation of momentum is a direct consequence of Newton's second law, explaining why momentum is conserved in both elastic and inelastic collisions in this experiment.

6.7. COEFFICIENT OF RESTITUTION FOR ELASTIC COLLISIONS OF SLIDERS WITH DIFFERENT MASSES

For the elastic collisions involving sliders of different masses, the coefficient of restitution was calculated using the results from Table 2. The coefficient of restitution e is defined as the ratio of the relative velocity of separation to the relative velocity of approach. Using the formula:

$$e = \frac{V_2 - V_1}{U_1 - U_2}$$



where:

- U_1 and U_2 are the initial velocities of sliders 1 and 2,
- V_1 and V_2 are the final velocities of sliders 1 and 2 after the collision.

For each case, the coefficient of restitution was determined and the average value was calculated. The average experimental coefficient of restitution was found to be close to the theoretical prediction derived in Exercise 5. Although slight deviations were present, likely due to air resistance and imperfect collisions, the experimental results align reasonably well with the theoretical expectation, confirming the elastic nature of the collisions.

6.8. PERCENTAGE OF KINETIC ENERGY LOST IN INELASTIC COLLISIONS

For the inelastic collisions, the average percentage of kinetic energy lost was calculated using the results from Table 5. The percentage of energy lost was derived from the initial and final kinetic energies of the system:

$$\% \text{ Energy Lost} = \frac{KE_{\text{initial}} - KE_{\text{final}}}{KE_{\text{initial}}} \times 100\%$$

where:

- KE_{initial} is the total initial kinetic energy before the collision,
- KE_{final} is the total final kinetic energy after the collision.

The average experimental percentage of kinetic energy lost was compared with the theoretical prediction from Exercise 4, which states that some kinetic energy is lost to heat, sound and deformation in inelastic collisions. The experimental result showed a higher percentage of energy loss compared to the theoretical one, likely due to additional factors such as friction between the sliders and imperfections in the collision.

6.9. EQUATION (1) FOR TWO OR THREE DIMENSIONAL COLLISIONS

In one dimensional collisions, the conservation of momentum is given by:

$$M_1U_1 + M_2U_2 = M_1V_1 + M_2V_2$$

For two or three dimensional collisions, the equation must account for the momentum components in each direction. The general form of the conservation



of momentum for two or three dimensions would be expressed as vector equations, such that:

$$\vec{P}_{\text{initial}} = \vec{P}_{\text{final}}$$

where:

$$M_1\vec{U}_1 + M_2\vec{U}_2 = M_1\vec{V}_1 + M_2\vec{V}_2$$

Here, $\vec{U}_1$ and $\vec{U}_2$ are the initial velocity vectors and $\vec{V}_1$ and $\vec{V}_2$ are the final velocity vectors. Each velocity vector would have components along the x, y and (for 3D) z axes. Momentum conservation would need to be applied separately for each axis, ensuring that the total momentum along each axis is conserved.

7. CONCLUSION

The experiment successfully demonstrated the conservation of momentum in both elastic and inelastic collisions using sliders on a nearly frictionless air track. For elastic collisions, momentum was conserved, and the transfer of kinetic energy was nearly ideal, with minimal losses due to air resistance and experimental imperfections. The coefficient of restitution for elastic collisions was close to the theoretical value, further supporting the findings.

In inelastic collisions, while momentum was still conserved, a significant amount of kinetic energy was lost, primarily due to friction and the conversion of energy into other forms, such as heat and sound. The calculated percentage of energy loss in inelastic collisions was notably higher than that in elastic collisions, consistent with theoretical predictions.

The experiment reinforced the principle that momentum conservation is independent of energy conservation, as total momentum remained constant in both types of collisions, even when kinetic energy was not conserved. Overall, the results were in good agreement with theoretical expectations, with minor deviations attributed to experimental conditions such as air resistance, noise, and imperfections in the collisions.



8. REFERENCES

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